Synoptic Question 30: The Physics of an Automatic Washing Machine

Reading Passage

Read the following article and then answer the questions that follow.

An automatic washing machine uses a combination of mechanical action, thermal energy, and chemical agents (detergents) to clean clothes. A key component is a rotating drum, usually mounted horizontally or slightly tilted. Clothes are placed inside the drum, which is then partially filled with water.

An electric motor drives the drum's rotation. During the wash cycle, the drum rotates back and forth at a relatively low speed, tumbling the clothes through the water. For the spin cycle, the drum rotates at a very high speed. The resulting centripetal force pushes the clothes against the perforated drum wall, while the water passes through the holes and is pumped out, effectively drying the clothes.

Many washing machines also include an electrical heating element submerged in the water. This element, essentially a large resistor, converts electrical energy into thermal energy to heat the water to a desired temperature, which improves the effectiveness of the detergent. The power consumption of the machine is dominated by the heating element and the motor, particularly during the high-speed spin cycle.

Modern machines often use electronic controls to manage the cycle timing, water level, temperature, and motor speed. The motor itself is typically an AC induction motor or a brushless DC motor, chosen for its efficiency and ability to provide variable speeds.

Data:

Data for a typical front-loading washing machine:

Table 1: Data for Washing Machine Analysis

Parameter	**Value**
Inner radius of drum, r	0.25 m
Length of drum, L	0.30 m
Mass of empty drum, m_{drum}	8.0 kg
Maximum mass of dry clothes, m_{load}	7.0 kg
Volume of water used in wash cycle	50 litres
Density of water, ρ_{water}	1000 kg m^{-3}
Specific heat capacity of water, c_{water}	4200 J kg^{-1} K^{-1}
Rotational speed during wash cycle	50 rpm
Rotational speed during spin cycle, $\omega_{spin,rpm}$	1200 rpm
Time taken to accelerate to spin speed, t_{accel}	4.0 s
Power rating of heating element, P_{heat}	2100 W
Mains voltage supply (RMS), V_{rms}	230 V
Efficiency of motor during spin, η_{motor}	75%

Questions

(a) During the wash cycle, water is heated.

(i) Calculate the mass of water used in the wash cycle. (1 litre = 1×10^{-3} m^3).

(ii) Calculate the energy required to heat this water from 15.0 °C to 40.0 °C.

(iii) Assuming the heating element is the only component drawing power, calculate the time taken to heat the water.

(iv) Calculate the resistance of the heating element.

(b) During the spin cycle, the drum rotates at high speed. Assume the clothes (m_{load}) stick uniformly to the inner wall of the drum.

(i) Convert the spin speed of 1200 rpm into angular velocity ω in rad s^{-1}.

(ii) Calculate the linear speed v of the clothes during the spin cycle.

(iii) Calculate the magnitude of the centripetal force acting on a small 0.10 kg piece of clothing during the spin.

(iv) Explain qualitatively why the water is extracted from the clothes during the spin cycle.

(c) Consider the acceleration phase of the spin cycle. The total moment of inertia I_{total} of the drum and wet clothes can be approximated as $I_{total} \approx (m_{drum} + m_{wet_load})r^2$. Assume the wet load mass is double the dry load mass.

(i) Estimate the moment of inertia I_{total} of the loaded drum.

(ii) Assuming constant angular acceleration, calculate the angular acceleration α as the drum speeds up.

(iii) Calculate the net torque τ_{net} required to produce this acceleration.

(iv) Calculate the work done W by this net torque during the acceleration phase.

(d) The motor provides the torque to accelerate and spin the drum.

(i) Calculate the maximum rotational kinetic energy of the loaded drum during the spin cycle.

(ii) Assuming the work done in (c)(iv) equals the final rotational KE, calculate the average mechanical power P_{mech} supplied by the motor during acceleration.

(iii) Calculate the average electrical power P_{elec} consumed by the motor during acceleration.

(e) Briefly explain the basic principle (interaction between magnetic fields and currents) by which an electric motor produces torque.

Solutions

(a) (i) Volume $V = 50$ litres $= 50 \times 10^{-3}$ m^3.
Mass $m_{water} = \rho_{water} V = (1000)(50 \times 10^{-3}) = 50.0$ kg.

(ii) $\Delta T = 40.0 - 15.0 = 25.0$ K.
Energy $Q = m_{water} c_{water} \Delta T = (50.0)(4200)(25.0) = 5.25 \times 10^6$ J.

(iii) Power $P_{heat} = 2100$ W.
Time $t = \frac{Q}{P_{heat}} = \frac{5.25 \times 10^6}{2100} = 2500$ s (approx 41.7 minutes).

(iv) $P = \frac{V_{rms}^2}{R} \implies R = \frac{V_{rms}^2}{P_{heat}}$
$R = \frac{(230)^2}{2100} = 25.19\ \Omega \approx 25.2\ \Omega$.

(b) (i) $\omega = 1200 \frac{\text{rev}}{\text{min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 125.7$ rad s^{-1}.

(ii) $v = r\omega = (0.25)(125.7) = 31.43$ m s$^{-1} \approx 31.4$ m s^{-1}.

(iii) $m_{piece} = 0.10$ kg.
$F_c = m_{piece} r \omega^2 = (0.10)(0.25)(125.7)^2 = 395$ N.
(Alternatively: $F_c = m_{piece} v^2 / r = (0.10)(31.43)^2/0.25 = 395$ N).

(iv) The drum wall provides the large inward centripetal force required to keep the clothes moving in a circle. Water droplets, being mobile, are not held as strongly by the fabric. Their inertia means they tend to continue moving in a straight line (tangential to the circle). They pass through the perforations in the drum and are collected and pumped away.

(c) $m_{wet_load} = 2 \times m_{load} = 2 \times 7.0 = 14.0$ kg.

(i) $I_{total} \approx (m_{drum} + m_{wet_load}) r^2 = (8.0 + 14.0)(0.25)^2 = (22.0)(0.0625) = 1.375$ kg m^2.

(ii) $\omega_f = 125.7$ rad s^{-1}, $\omega_i = 0$.
$\alpha = \frac{\omega_f - \omega_i}{t_{accel}} = \frac{125.7 - 0}{4.0} = 31.4$ rad s^{-2}.

(iii) $\tau_{net} = I_{total} \alpha = (1.375)(31.4) = 43.175$ Nm ≈ 43.2 Nm.

(iv) Angular displacement $\theta = \omega_i t + \frac{1}{2}\alpha t^2 = 0 + \frac{1}{2}(31.4)(4.0)^2 = 251.2$ rad.
Work done $W = \tau_{net}\theta = (43.175)(251.2) = 10845$ J $\approx 1.08 \times 10^4$ J.
(Alternatively: $W = \Delta KE_{rot} = \frac{1}{2} I \omega_f^2 = \frac{1}{2}(1.375)(125.7)^2 = 10857$ J). Let's use 10.9 kJ.

(d) (i) $KE_{rot,max} = \frac{1}{2} I_{total} \omega_{spin}^2 = \frac{1}{2}(1.375)(125.7)^2 = 10857$ J ≈ 10.9 kJ.

(ii) $P_{mech,avg} = \frac{W}{t_{accel}} = \frac{10857}{4.0} = 2714$ W ≈ 2.71 kW.

(iii) $\eta_{motor} = \frac{P_{mech}}{P_{elec}} \implies P_{elec} = \frac{P_{mech}}{\eta_{motor}}$
$P_{elec,avg} = \frac{2714}{0.75} = 3619$ W ≈ 3.62 kW.

(e) An electric motor typically uses coils carrying an electric current placed within a magnetic field. The magnetic field exerts a force (the Lorentz force, $F = BIL \sin\theta$) on the current-carrying wires. These forces create a turning effect, or torque, on the coil assembly (rotor), causing it to rotate and drive the drum.